

Per-channel changes account for most long-term decoder drift in people with paralysis

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Abstract

Implanted brain–computer interfaces can restore communication and movement after paralysis, but their decoders lose accuracy as recordings change. What changes, and what a correction must fix, remains unclear. Here, we show that in people, most of this loss can be undone by re-learning one weight per recording channel. We analysed public long-term recordings from three monkeys and 14 people with one pipeline. In all three people with decoding data, this re-weighting recovered most of the accuracy lost over days to more than a year, for linear and recurrent decoders. In monkeys it recovered much less, even when they decoded the same variable. Decay rates differed about a hundredfold between individuals, and weak regularization exaggerated decay. Label-free alignment of neural activity added nothing beyond rescaling each channel, and brief electrode silences were mostly session-to-session fluctuation. These results show what recalibration in clinical systems has to correct, and set baselines for new methods.

Keywords: brain–computer interface, intracortical recording, Utah array, neural drift, decoder recalibration, electrode stability

¹ Introduction

² Intracortical brain–computer interfaces (BCIs) record from small populations of neu-
³ rons and turn their activity into commands. They have allowed people with paralysis

4 to move robotic arms and computer cursors¹⁻⁴, to type by attempted handwriting⁵
5 and to speak through a decoded voice⁶⁻⁸. Recent work describes daily home use over
6 years⁹. For these devices to become everyday tools, their decoders must stay accurate
7 with little effort from the user.

8 The recorded signals are not stable. Within a day, signals shift as electrodes
9 move or neurons change their firing¹⁰. Across days, the activity seen by each elec-
10 trode changes¹¹⁻¹³. Over months and years, arrays gradually lose the ability to record
11 spikes¹⁴⁻¹⁷. Most systems therefore recalibrate their decoders often, which takes time
12 away from the user¹⁸. Many methods aim to reduce this cost. Decoders can be trained
13 on data from many days¹⁹ or updated during use^{18,20-22}. Neural activity can be
14 aligned to a reference day without new labels²³⁻²⁶, and monitoring scores have been
15 proposed to signal when recalibration is needed²⁷.

16 Choosing among these approaches requires knowing what actually changes in the
17 recordings. If the decoded signal moves to different channels, a decoder must be rebuilt.
18 If each channel keeps its tuning but changes in strength, a much smaller correction
19 may suffice. Earlier work modelled per-electrode drift as baseline shifts in monkeys
20 over weeks^{13,18}, and human decoders absorb day-to-day changes with a full input layer
21 learned for each day^{5,28}. Other studies describe drift as rotation of preferred directions
22 or of decoder weights^{22,27}. Whether a strictly per-channel correction suffices, and over
23 what span of time, has not been isolated, and most comparisons used one dataset,
24 weeks of data and different baselines²⁹. On the hardware side, long-term studies report
25 falling array yield and impedance^{14,15,17,30}, but how single electrodes come and go is
26 less well described.

27 Public long-term datasets now allow these questions to be asked across individuals
28 with one method. They include three years of finger decoding in one monkey³¹ and
29 reaching recordings from two monkeys in another laboratory^{24,32}. They also include
30 closed-loop cursor sessions spanning two to seven years in three BrainGate partici-
31 pants, array recordings from 20 arrays in 14 participants¹⁷, and closed-loop sessions
32 with fixed decoders²⁷.

33 Here, we analysed these datasets with a single pipeline. We trained decoders on
34 one day and tested them on later days after a ladder of increasingly powerful cor-
35 rections. Decay rates differed about a hundredfold between individuals, and weakly
36 regularized decoders exaggerated decay. In all three people with decoding data, re-
37 learning one weight per channel restored most of the lost accuracy at every gap tested,
38 for linear and recurrent decoders. In monkeys, this correction recovered much less at
39 long gaps, even when we decoded the same variable as in humans. Label-free align-
40 ment added nothing beyond rescaling each channel, and statistics of neural change
41 predicted decoder accuracy little better than elapsed time. At the electrode level,
42 brief silences were mostly session-to-session fluctuation, whereas silences lasting over
43 a month were real and often reversed. We release the code, a calibrated simulator and
44 all derived results.

Results

A common pipeline for six decoding datasets

We assembled four public datasets (Fig. 1a and Supplementary Table 1). Decoder analyses used six individuals. Monkey N was recorded over 1,242 days with Utah arrays in motor cortex during a two-finger task^{31,33}. Monkeys C and M were recorded over three and 1.5 years during reaching^{24,32}. BrainGate participants T6, T5 and T9 contributed 124, 92 and 117 closed-loop cursor sessions over 3.1, 7.2 and 2.0 years¹⁷. Electrode analyses used all 20 arrays from 14 BrainGate participants (2,289 sessions) and monkey N. Monitoring analyses used closed-loop sessions in which participants T11 and T5 controlled a cursor with a fixed decoder²⁷.

For each individual, we trained a linear decoder on one session and tested it on later sessions at gaps of 1 to 480 days. Before testing, every channel of the later session was z-scored with that session’s own mean and standard deviation. This step, which we call renormalization, needs no labels; the result is the renormalized fixed decoder. We then applied a ladder of corrections (Fig. 1b and Methods). Label-free rungs updated channel means or aligned neural activity to the training day. Supervised rungs used labelled trials from the later day. They either re-learned one weight per channel in front of the frozen decoder, or refitted the decoder while shrinking it toward the old one. A decoder trained on the later day served as the reference. We report retention, the ratio of a decoder’s accuracy (coefficient of determination, R^2) to that of the reference on the same test trials.

Decoders decay at rates that differ a hundredfold

Without any correction, decoders failed: reusing the training day’s normalization gave median R^2 below zero at most gaps (Supplementary Table 2). We therefore treat the renormalized fixed decoder as the baseline.

How much accuracy survives a single night differed between species (Fig. 2a,b). For true one-day gaps, retention was 0.74 in monkey N (95% confidence interval 0.62–0.83; $n = 21$ pairs), 0.70 in monkey C (0.58–0.77; $n = 31$) and 0.80 in monkey M (0.70–0.86; $n = 12$). In participant T9 it was 0.58 (0.33–0.76; $n = 19$). One-day pairs were rare in T6 ($n = 5$; retention 0.41, interval -2.45 – 0.99) and T5 ($n = 2$). At two days, retention was 0.79 in T6 ($n = 19$) and 0.59 in T5 ($n = 15$). Across the six individuals, retention after one to two days differed (Kruskal–Wallis test, $H = 11.4$, d.f. = 5, $P = 0.045$; one value per training session).

Longer-term decay differed far more. Retention fell to one half within 1.1 days in T9, 3.7 days in T5, 8.1 days in monkey N and 9.2 days in monkey C. It took 34 days in monkey M and 125 days in T6 (Fig. 2a,c). The reference decoder itself stayed equally accurate across gaps (Supplementary Table 2), so the decline reflects a growing mismatch between the decoder and the day, not poorer recordings.

Weak regularization exaggerates decay

Part of the measured decay came from the decoder rather than the recordings. With a ridge penalty of $\alpha = 10^4$ instead of the small values used in the original dataset

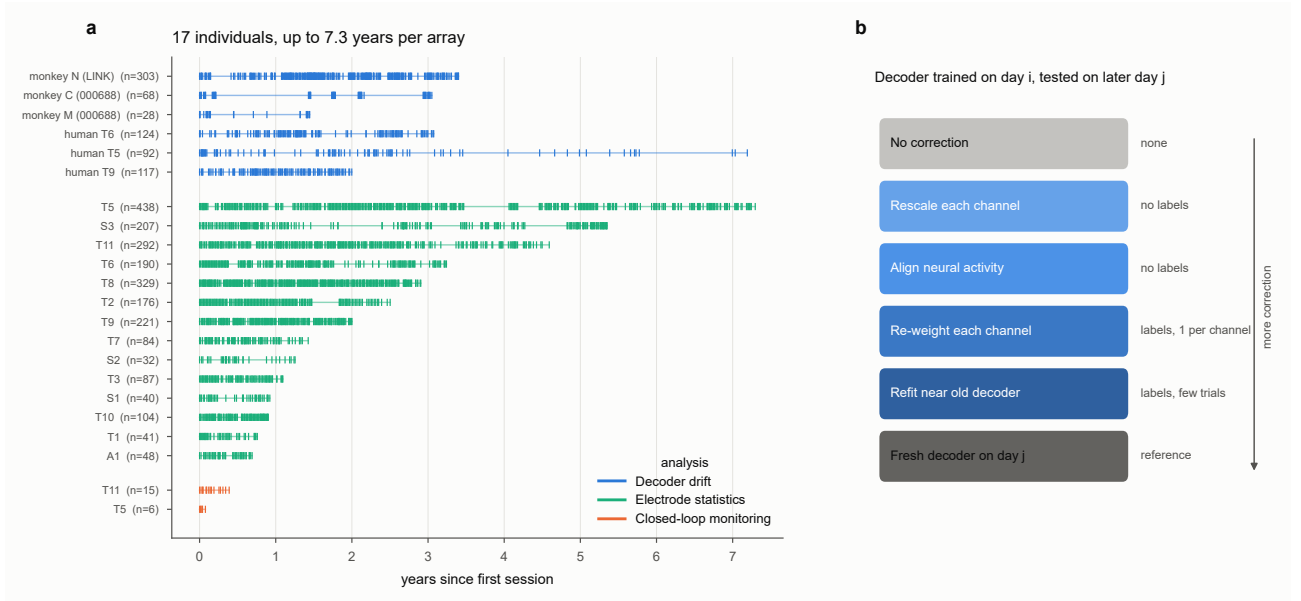


Fig. 1 Data and analysis design. **a**, Sessions used in this study, by individual and analysis, against years since each individual's first session. Each tick is one session. Blue, decoder drift (monkeys N, C and M; human participants T6, T5 and T9); green, electrode statistics (20 arrays in 14 BrainGate participants); orange, closed-loop monitoring (T11 and T5). Counts are per analysis; some human sessions enter more than one analysis. **b**, The correction ladder. A decoder trained on day i is tested on a later day j after increasingly powerful corrections: none; label-free corrections (rescaling each channel, aligning neural activity); and corrections that use labelled trials from day j (re-learning one weight per channel; refitting the decoder while shrinking it toward the old one). A decoder trained on day j is the reference.

papers³¹, the time to half retention rose from 8.1 to 16 days in monkey N and from 125 to 229 days in T6 (Fig. 2c,d). It changed little in monkeys C (9.2 to 10 days) and M (34 to 37 days), whose original decoders were already more strongly regularized, and in T5 (3.7 to 7.9 days) and T9 (1.1 to 1.1 days). Per training session, absolute R^2 on later days rose by a median of 0.061 in monkey N (two-sided Wilcoxon signed-rank test, $P = 8.2 \times 10^{-15}$) and 0.019 in T6 ($P = 9.2 \times 10^{-22}$).

A simple rule captures this benefit using training-day data only: choose the largest penalty whose same-day accuracy stays within 1% of the best (Fig. 5a,b and Supplementary Table 3). Compared with $\alpha = 0.1$, it raised cross-day R^2 in every individual, with session-level median gains of 0.060 in monkey N (40 of 40 sessions; $P = 1.8 \times 10^{-12}$), 0.021 in T6 ($P = 1.8 \times 10^{-12}$), 0.033 in T5 ($P = 1.8 \times 10^{-12}$) and 0.045 in T9 ($P = 1.8 \times 10^{-12}$). Gains were small in monkeys C (0.017) and M (0.001). Same-day accuracy did not fall (median change 0.000 to +0.008). Ordinary same-day tuning chose similar penalties in most individuals (Methods), so the lesson is that small default penalties exaggerate drift, rather than that cross-validation fails.

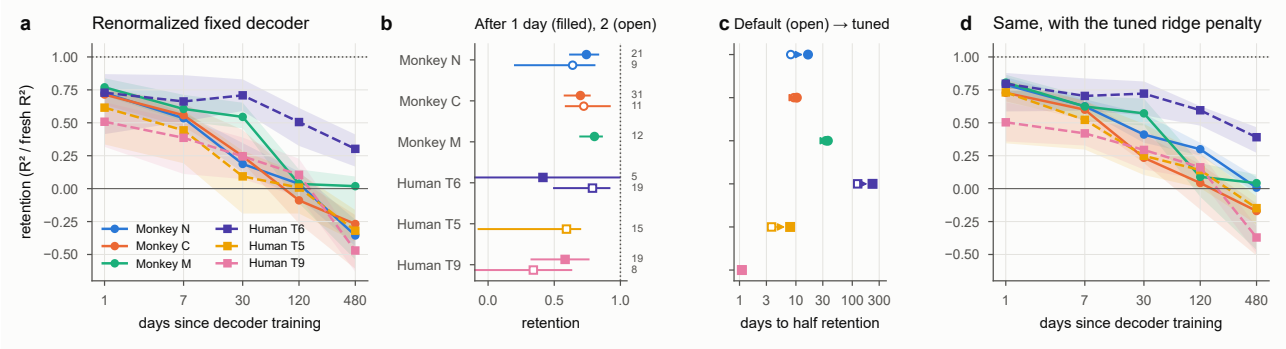


Fig. 2 Decoders decay at rates that differ between individuals, and weak regularization exaggerates decay. **a**, Retention of the renormalized fixed decoder (decoding R^2 divided by that of a decoder trained on the test day) against days since training, with the ridge penalty used in the original dataset papers. Lines show medians and bands show 95% confidence intervals from a cluster bootstrap over training sessions. Circles and solid lines, monkeys; squares and dashed lines, humans. Blue, monkey N; orange, monkey C; green, monkey M; purple, T6; amber, T5; pink, T9. **b**, Retention after exactly one day (filled) and two days (open), with 95% confidence intervals; numbers give pairs. Markers are omitted when fewer than three pairs were available. **c**, Days until retention falls to one half with the default penalty (open) and with the tuned penalty (filled; logarithmic axis). **d**, As in **a**, for decoders trained with the tuned penalty ($\alpha = 10^4$). Pair counts are given in Supplementary Table 2.

Re-weighting each channel restores most accuracy in people

We next asked what kind of change the remaining loss reflects. Re-learning one weight per channel in front of the frozen decoder lets each channel's contribution grow, shrink or vanish, but keeps the decoder's pattern across time lags and outputs. In all three people, this correction restored most of the lost accuracy at every gap (Fig. 3a). It recovered 64–81% of the loss in T6, 72–85% in T5 and 76–84% in T9 (medians per gap; Fig. 3b). Retention after re-weighting stayed between 0.70 and 0.96 even 480 days after training. In the monkeys, the same correction recovered 14–34% of the loss in monkey N, 28–42% in C and 19–35% in M, and retention after re-weighting fell to 0.10–0.22 at 480 days.

The human result did not depend on the decoder. A recurrent network decoder (Methods), regularized to avoid memorizing the closed-loop sessions, was as accurate as the linear decoder in people (median R^2 0.24, 0.49 and 0.20 versus 0.25, 0.50 and 0.31). Re-weighting its input channels recovered 75–102%, 82–92% and 83–105% of its loss (Fig. 3b). In monkeys the network was more accurate than the linear decoder (median R^2 higher by 0.11, 0.33 and 0.42), but re-weighting again recovered less at long gaps (Supplementary Table 4).

Could the species difference reflect what was decoded? Human decoders predicted a two-dimensional direction, whereas monkey decoders predicted four kinematic signals. We therefore rebuilt the reaching monkeys' sessions to decode the direction of hand movement only, mirroring the human target (Methods). Re-weighting then recovered 40–58% of the loss in monkey C and 31–46% in monkey M across gaps (Fig. 3c). This was more than with kinematic targets but still well below the human values. The decoded variable explains part of the difference, but not most of it.

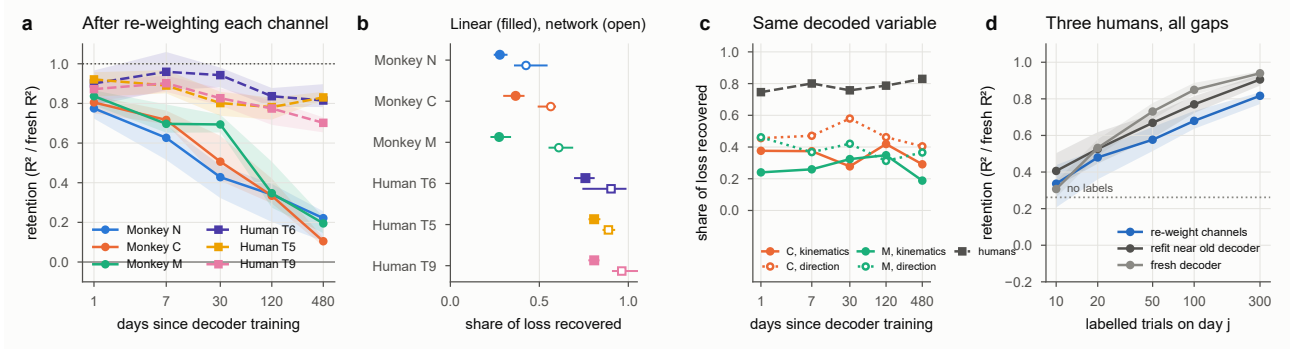


Fig. 3 Re-weighting each channel restores most accuracy in people. **a**, Retention after re-learning one weight per channel in front of the frozen linear decoder, with up to 300 labelled trials from the test day. Lines, bands and colours as in Fig. 2a. **b**, Share of the loss recovered by re-weighting, $(\text{re-weighted } R^2 \text{ minus renormalized } R^2) / (\text{reference } R^2 \text{ minus renormalized } R^2)$, for the linear decoder (filled) and a recurrent network decoder (open), median over all pairs with 95% cluster-bootstrap confidence intervals. **c**, Share of the loss recovered in the reaching monkeys when decoding hand kinematics (solid) or hand-movement direction only (dotted), the variable decoded in the human sessions, with the human median for comparison (grey). **d**, Retention against the number of labelled trials on the test day, pooled over the three human participants and all gaps, for re-weighting channels, refitting the decoder while shrinking it toward the old one, and training a fresh decoder. The dotted line marks the renormalized fixed decoder, which uses no labels.

Per-channel changes also reconcile our result with reports that decoders rotate across days^{22,27}. Those reports measured the angle between decoders fitted on different days, allowing only one global change of scale. When we also allowed one scale per channel, the angle between decoders fitted on different days fell by 9–34° in every individual (Supplementary Table 5). Angles that allow only a global scale therefore overstate how much decoders rotate. Decoder weights were too variable, however, to separate the species on this measure, so we rely on the functional test above.

Re-weighting is a diagnostic of what changes, not necessarily the most efficient way to recalibrate. Pooled over the three participants and all gaps, retention after re-weighting was 0.48 with 20 labelled trials, 0.58 with 50 and 0.82 with 300, against 0.26 without labels (Fig. 3d). With 50 trials, refitting the decoder while shrinking it toward the old one reached 0.67, and a fresh, strongly regularized decoder reached 0.73. Trials lasted about 1.6–3.9 s, so 50 trials take one to three minutes. In these participants, then, a short block of labelled data restored most of the accuracy with any well-regularized refit.

Rescaling each channel is the label-free step that matters

Several methods realign neural activity to a reference day without labels^{23–26}. Their benefits were usually measured against a fixed decoder or against one whose channel means alone were updated^{22,25}. We compared four label-free corrections with renormalization, which rescales each channel (Fig. 4a and Supplementary Table 6).

Updating channel means alone was worse than renormalization in every individual (Fig. 4a); in monkey N the difference was 0.061 in R^2 (two-sided Wilcoxon signed-rank test over training sessions, $P = 3.0 \times 10^{-11}$). Rotating the dominant neural subspace

148 onto the training day by orthogonal Procrustes analysis³⁴ did not close this gap.
149 Applied after updating means, it lowered R^2 further in monkey N (by -0.063 ; $P =$
150 3.4×10^{-8}). Applied after renormalization, it lowered R^2 slightly in every individual,
151 by -0.019 in monkey N ($P = 6.6 \times 10^{-9}$) and by -0.011 , -0.026 and -0.063 in T6,
152 T5 and T9. Matching the full channel covariance (CORAL³⁵) improved slightly on
153 renormalization in monkey N (by 0.011 ; $P = 1.9 \times 10^{-9}$) but not in the other five
154 individuals. A faithful implementation of the factor-analysis stabilizer of Degenhart
155 and colleagues²³ was better than renormalization in T5 (by 0.070 ; $P = 4.8 \times 10^{-9}$)
156 and worse in the other five individuals. That method was designed for closed-loop use
157 with a latent decoder, and offline tests may understate it. In these data, however, the
158 label-free step that mattered was rescaling each channel, and the gains reported for
159 alignment methods over mean-only baselines²² largely reflect this step.

160 Neural change adds little beyond time since calibration

161 A monitor that warns when a decoder is failing could make recalibration less fre-
162 quent. MINDFUL scores instability as the divergence between distributions of neural
163 features, or of decoder outputs, on a test day and a reference day²⁷. Such scores cor-
164 relate with decoder error, but both grow with time since calibration, so a correlation
165 alone does not show that a score adds information.

166 Offline in monkey N (537 session pairs), elapsed time alone correlated with decoder
167 accuracy (Spearman $|\rho| = 0.67$; Fig. 4b). Neural divergence correlated more weakly
168 ($|\rho| = 0.34$), and its partial correlation fell to 0.16 after controlling for elapsed time.
169 Output divergence behaved the same way ($|\rho| = 0.22$, partial 0.16). Adding both
170 divergences to elapsed time slightly increased the cross-validated error of predicted R^2
171 (0.090 versus 0.085). A broader set of label-free statistics fared no better (Supplemen-
172 tary Note 1), although day-to-day performance beyond the time trend was reliable
173 (split-half reliability 0.84).

174 We then re-analysed the public MINDFUL data, in which T11 and T5 controlled
175 a cursor in closed loop with a fixed decoder (Fig. 4c and Supplementary Table 7). In
176 T11 (17.1 days), neural divergence correlated with angular error ($r = 0.82$), consistent
177 with the original report²⁷. Most of this association was shared with elapsed time
178 (partial $r = 0.37$), and adding neural divergence did not improve leave-one-day-out
179 prediction of error (17.5° versus 17.1°). Output divergence kept a partial correlation
180 of 0.73 , and adding it reduced the error to 11.9° . In T5, error was barely related to
181 elapsed time over 32.9 days ($r = 0.17$), and both divergences tracked error (partial
182 $r = 0.49$ for neural and 0.78 for output divergence). These comparisons involve few
183 days, and output divergence and angular error are both computed from the cursor's
184 motion, so part of their link is mechanical. Within these limits, elapsed time was as
185 good a guide as neural statistics in most comparisons.

186 Shrinking toward the old decoder makes small recalibrations 187 safe

188 When labelled data are scarce, refitting a decoder can make it worse. In monkey N
189 (86 session pairs), we refitted the decoder by ridge regression shrunk toward the old

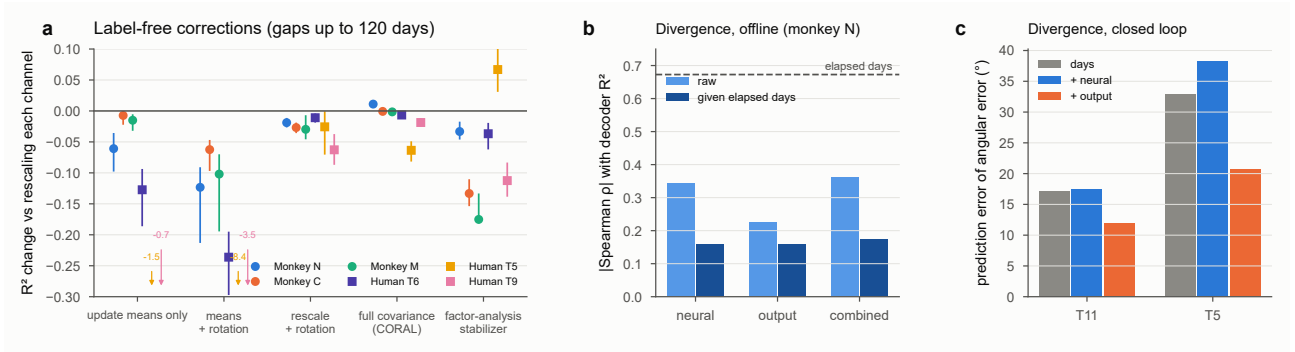


Fig. 4 Label-free corrections and monitoring add little to rescaling and elapsed time.
a, Change in decoding R^2 relative to renormalization (rescaling each channel) for five label-free corrections, per individual, averaged over each training session's pairs at gaps up to 120 days. Points show medians over training sessions and bars show 95% cluster-bootstrap confidence intervals. Colours and markers as in Fig. 2. **b**, Offline monitoring in monkey N (537 pairs): absolute Spearman correlation between decoder R^2 and the divergence of neural features, decoder outputs or both, before (light) and after (dark) controlling for elapsed days. The dashed line marks elapsed days alone. **c**, Closed-loop monitoring in the public MINDFUL data: leave-one-day-out mean absolute error in predicting angular error from elapsed days (grey), with neural divergence added (blue) or with output divergence added (orange).

decoder, penalizing changes in both the weights and the intercept^{20,21,36} (Fig. 5c and Supplementary Table 8). Choosing the shrinkage strength by cross-validation on a few trials was unreliable: with 10 trials, 10.5% of recalibrations ended worse than no recalibration (Fig. 5d). A fixed strong shrinkage, the value that worked best on pairs from other training sessions, lowered this to 1.2% (exact McNemar test on paired outcomes, $P = 0.008$; Holm-adjusted over five data budgets, $P = 0.039$). With 20 trials the rates were 8.1% and 0.0% ($P = 0.016$; adjusted $P = 0.062$). With 50 or more trials, and at gaps of 30 days or more, refitting from scratch did as well as the shrunk refit.

Brief electrode silences are fluctuation, long ones are real

We next asked how single electrodes change over years. We used the BrainGate yield data, which report threshold-crossing rates and impedance for every electrode in every session¹⁷, and the matching data from monkey N. An electrode was active when its threshold-crossing rate exceeded 2 Hz, the definition used by BrainGate¹⁷. Array yield and impedance declined over years in both species (Fig. 6a,b). Impedance fell in 17 of 18 human arrays with measurements (median Spearman ρ with time -0.92) and in monkey N, from a median of 302 k Ω in the first year to 171 k Ω in the fourth ($\rho = -0.88$), replicating earlier reports^{14–17,30}.

Electrodes often fell silent and later recovered, but most such events were fluctuation rather than failure and repair (Fig. 6c and Supplementary Table 9). Of 730 initially active human electrodes that stayed below threshold for at least three consecutive sessions, 80% later recovered for at least three sessions. The same statistic computed after randomly shuffling each electrode's session order, which destroys any

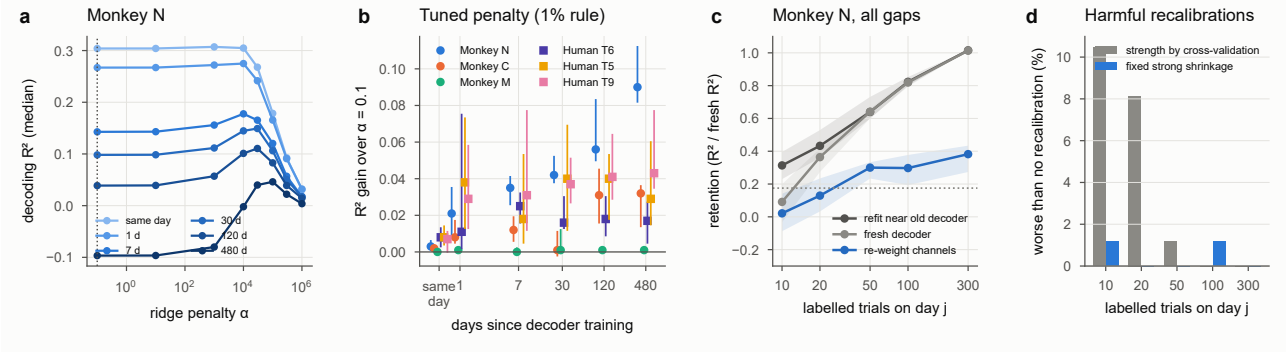


Fig. 5 Simple recipes for robust decoders. **a**, Median decoding R^2 in monkey N against the ridge penalty α , on held-out trials of the training day (lightest) and on sessions 1, 7, 30, 120 and 480 days later (darker). The dotted line marks $\alpha = 0.1$, the value used in the original dataset paper. **b**, Gain in R^2 from the 1% rule (largest α whose same-day R^2 is within 1% of the best) over $\alpha = 0.1$, on the same day and at each gap, per individual; medians with 95% cluster-bootstrap confidence intervals. **c**, Retention in monkey N against the number of labelled trials on the test day, pooled over gaps, for refitting the decoder while shrinking it toward the old one, training a fresh decoder and re-weighting channels. **d**, Percentage of recalibrations (86 pairs) that ended more than 0.01 R^2 below the renormalized fixed decoder, with the shrinkage strength chosen by cross-validation on the available trials (grey) or fixed at the value that worked best on other training sessions (blue).

temporal process, gave 79%. The match held when the threshold was fixed in microvolts for each electrode, so that changes in noise level could not move it (72% versus 72%), and when silence required rates below 0.5 Hz (71% versus 71%). Silences lasting at least 30 days were different. They were twice as common as in the shuffled data (168 versus 81), and 57% of them ended in recovery, more than in the shuffled data (41%; Fisher's exact test, $P = 0.021$). Recovery here means the electrode again recorded spiking, not necessarily from the same neuron.

Spiking declined somewhat faster at the edge of the array than in its interior (Fig. 6d). Among initially active electrodes, the edge decline was steeper in 13 of 17 human arrays with at least 10 such electrodes (one-sided Wilcoxon signed-rank test with the array as the unit, $P = 0.036$), and decline correlated with distance from the array centre ($P = 0.013$). After adjusting for each electrode's initial firing rate, the edge effect weakened (12 of 17 arrays; $P = 0.087$). In monkey N, edge electrodes declined faster within both arrays for threshold crossings but not for spike-band power. These trends agree with models in which micromotion strains tissue most at the array edges³⁷, but they are modest and partly explained by initial activity.

A calibrated simulator predicts untested corrections

Methods for stable decoding are hard to compare without a common test bed²⁹. We built a feature-level simulator that takes a real session and returns it as it might appear days to years later (Methods). It combines mixing of signals between channels, gradual turnover of each channel's tuning and switching of electrodes between active and silent states. Switching rates came from monkey N's electrode statistics, and the mixing and turnover parameters were calibrated so that the simulated correction



Fig. 6 Brief electrode silences are fluctuation; long silences are real. **a**, Percentage of electrodes with spiking activity (threshold-crossing rate of at least 2 Hz) against years since implant, for 20 human arrays (purple) and the two arrays of monkey N (blue). **b**, Median impedance of each array relative to its first sessions (logarithmic axis). **c**, Percentage of silent episodes in initially active human electrodes that ended in recovery (at least three sessions above 4 Hz), observed (purple) and after shuffling each electrode's session order (grey), for four definitions of silence: at least three sessions below 2 Hz; the same with a fixed microvolt threshold per electrode; below 0.5 Hz; and lasting at least 30 days. Numbers give silent episodes. **d**, Difference in the decline of log threshold-crossing rate per year between edge and interior electrodes, per human array with at least 10 initially active electrodes, unadjusted (filled) and adjusted for initial firing rate (open). Negative values mean faster decline at the edge.

ladder matched monkey N's during its first 700 days (root-mean-square error 0.07 in retention; Fig. 7a).

We tested the simulator on the later days, which it had not seen, against two baselines (Fig. 7b). A simulator without drift was far worse (median absolute error 0.37–0.39). A stronger baseline simply reused the median retention measured in the first 700 days. For the corrections used in calibration, and for different amounts of labelled data, this empirical curve was as good as or better than the simulator (0.09 versus 0.12, and 0.07 versus 0.11). For corrections that the simulator was not fitted to, it was clearly better (0.06 versus 0.16). The simulator is therefore most useful for predicting how new corrections will behave. Real decoders decayed more slowly in the later years, and a calibration on that period gave weaker session-to-session mixing (0.77 versus 0.96) and slower turnover (time constant 527 versus 360 days; Fig. 7c and Supplementary Table 10), suggesting that drift slows as an implant ages in this monkey. Training decoders on simulated futures helped little (R^2 higher by 0.010) compared with stronger regularization (0.042) or training on several past sessions (0.056; Supplementary Table 11).

Discussion

We measured how decoders and electrodes change over years in three monkeys and 14 people with one pipeline. The central finding is that in people, most of the accuracy a decoder loses over days to more than a year is recovered by re-learning one weight per recording channel. This held in all three participants with decoding data and for

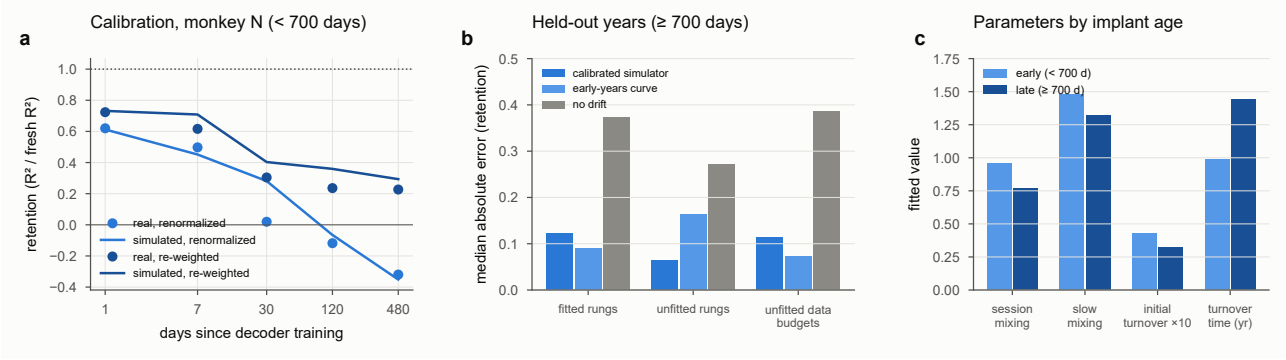


Fig. 7 A calibrated simulator predicts untested corrections. **a**, Calibration on monkey N sessions before day 700. Points show real median retention and lines show the calibrated simulator, for the renormalized fixed decoder and after re-weighting channels. **b**, Validation on sessions after day 700: median absolute error between predicted and real retention for corrections used in calibration, corrections not used in calibration and amounts of labelled data not used in calibration. Dark blue, calibrated simulator; light blue, the median retention measured before day 700; grey, the simulator with drift switched off. **c**, Parameters calibrated separately on early (< 700 days) and late (≥ 700 days) sessions: session-to-session mixing, slow mixing, initial turnover ($\times 10$) and turnover time constant (years).

linear and recurrent decoders. In monkeys, the same correction recovered much less at long gaps, and decoding the same variable as in the humans narrowed this difference only partly.

This result has a simple interpretation. In these human recordings, the decoder's pattern across time lags and outputs remained valid for each channel, while the strength with which each channel carried the decoded signal changed. It extends the observation that parameters on the same electrode change together between days¹³, which was made in monkeys over weeks and modelled as shifts in baseline, to multiplicative changes over years in people. It also suggests that the full input layers learned for each day in human decoders^{5,28} mainly absorb per-channel changes, and that part of the rotation reported for human decoders^{22,27} reflects per-channel scaling. Our results do not reveal the cause. Changes in the number or proximity of neurons near each electrode would act this way^{38,39}, as would changes in noise. Why monkeys differ is also open: species, cortical areas, tasks and recording features all differ between the datasets, and the human tasks were closed-loop.

What does this mean for recalibration? In these participants, a short block of labelled trials restored most of the accuracy with any well-regularized refit, so re-weighting is not needed for data efficiency. Its value is that it identifies the change that recalibration has to correct, and that it does so while keeping the decoder the user has learned to control, which may matter in closed loop. Label-free methods, by contrast, did not recover this change beyond rescaling each channel. Three further recipes follow from our results. Daily per-channel renormalization should be the baseline against which new methods are compared. Ridge penalties should be tuned rather than left at small defaults, because weak regularization exaggerates apparent drift. And when

281 recalibrating from a few trials, shrinking toward the previous decoder, including its
282 intercept, avoids recalibrations that do more harm than good.

283 Decay rates varied about a hundredfold between individuals, from one day to sev-
284 eral months to lose half of the accuracy. A fixed recalibration schedule will therefore
285 be wrong for many users. The monitoring analyses suggest that statistics of neural
286 change are a weak guide to when recalibration is needed, beyond elapsed time. Diver-
287 gence of the decoder’s output was more informative in closed loop, although part of
288 that link is mechanical. Monitoring studies should report correlations after controlling
289 for elapsed time and compare against a time-only baseline.

290 The electrode analyses refine common descriptions of array decline. Electrode
291 activity near the yield threshold fluctuated strongly from session to session, and most
292 short silences were no more than that. Yield counts from single sessions are there-
293 fore noisy, and decoders should not discard channels permanently after a few silent
294 sessions. Silences lasting over a month were real and still reversed in more than half
295 of cases. The edge effect we found is modest and partly explained by initial activity.
296 Hahn and colleagues analysed yield, decoding signal quality and tuning stability in
297 the same human data¹⁷. We add electrode-level analyses of silence and recovery with
298 a shuffled null, and a test of edge-biased spiking loss with the array as the unit.

299 Our study has limitations. All analyses were offline. In closed loop, users adapt to
300 their decoder⁴⁰, which could slow or hide decay. Human decoding labels were inferred
301 from the direction from cursor to target, and the online decoder shaped both the
302 neural activity and the cursor path. Three participants and three monkeys cannot
303 establish population rates, and the datasets differ in task, signal type, array placement
304 and implant age. The simulator is calibrated to feature-level statistics in one monkey
305 and does not model biophysics. Finally, we tested label-free methods offline, where
306 they may perform differently than in closed loop.

307 Long-term public datasets made this study possible. We release our code, derived
308 results and the calibrated simulator so that others can test methods against realistic
309 drift. Our results suggest that, in people, keeping a decoder working over months is
310 mainly a matter of tracking how strongly each channel carries the signal, and that
311 the corrections to compare against are simple ones.

312 Methods

313 Datasets and ethics

314 All data were public and de-identified. The original studies obtained ethical approval
315 and informed consent, as described in the cited publications. We collected no new
316 data.

317 *Monkey N* (LINK; DANDI Archive 001201⁴¹; ref.³¹). One rhesus macaque had
318 Utah arrays in the hand area of primary motor cortex³³. The release contains 312
319 sessions on 303 days over 1,242 days, in which the monkey moved two finger groups to
320 targets in centre-out or random-target blocks. We used the 96 released channels (64
321 on the medial array and 32 on the lateral array), with spike-band power and threshold
322 crossings at -4.5 times the root-mean-square noise in 20 ms bins⁴². Outputs were the

position and velocity of the two finger groups. Electrode impedance was stored on 228 days.

Monkeys C and M (DANDI Archive 000688⁴³; refs. ^{24,32}). Both monkeys had Utah arrays in motor and premotor cortex and performed centre-out and random-target reaching. Monkey C had 68 sessions from October 2013 to October 2016, and monkey M had 28 sessions from January 2014 to June 2015. We summed the spikes of all sorted units on each electrode in 20 ms bins, giving 96 (C) or 192 (M) channels. Hand position and velocity were interpolated to bin centres.

BrainGate participants (Dryad⁴⁴; ref. ¹⁷). The yield data contain one recording from each of 2,289 research sessions in 14 participants with 20 arrays (A1, S1–S3, T1–T3 and T5–T11). For every electrode, they list threshold-crossing rates at several thresholds (with the threshold in microvolts), impedance and grid position. Nineteen arrays were in the hand area of motor cortex and one was in the middle frontal gyrus. The decoding data contain closed-loop cursor sessions with spike-band power in 10 ms bins, cursor and target positions and the periods of attempted movement. We used all sessions of T6 (124 sessions over 3.1 years, one array, 96 channels), T5 (92 sessions over 7.2 years, two arrays, 192 channels) and T9 (117 sessions over 2.0 years, two arrays, 192 channels). Sessions contained a median of 367, 551 and 275 trials, with median movement periods of 1.6, 1.7 and 3.9 s.

MINDFUL (Dryad⁴⁵; ref. ²⁷). Closed-loop cursor blocks in which T11 (15 days) and T5 (6 days) used a fixed decoder, with neural features in 20 ms bins (spike-band power for T11 and threshold crossings for T5), the decoder’s velocity output and the angular error of the cursor.

Features, decoders and accuracy

For monkey N and the human participants we used spike-band power (human 10 ms bins averaged in pairs). For monkeys C and M we used per-electrode spike counts. Each channel was z-scored with the mean and standard deviation of the decoding session’s training segment, computed without labels (renormalization). The linear decoder was ridge regression⁴⁶ on 8 causal lags (160 ms) of the z-scored features, with an unpenalized intercept. The default penalty was $\alpha = 0.1$ for monkey N and the humans, as in ref. ³¹, and $\alpha = 1$ for monkeys C and M, whose spike-count features are sparser. Outputs were finger position and velocity for monkey N, hand position and velocity for monkeys C and M, and for the humans the intended movement direction, defined as the unit vector from cursor to target during attempted movement. Accuracy was the coefficient of determination R^2 , averaged over outputs and computed on held-out trials of the test session. In monkey N the first 300 of 375 trials trained the decoder, as in ref. ³¹; in the other datasets the first 80% of trials did.

The recurrent decoder was a long short-term memory network⁴⁷ with 64 hidden units, input dropout⁴⁸ of 0.5 and 20-bin input windows (400 ms). It was trained with AdamW⁴⁹ (learning rate 10^{-3} , weight decay 0.01) for up to 4,000 iterations of 256 samples, with early stopping on the last 20% of training trials checked every 100 iterations. Without these measures, a 256-unit network memorized the human training sessions (training R^2 0.97, test R^2 -1.1 on one session). The network ladder used 25 training sessions per individual chosen at random with a fixed seed.

Session pairs

For each training session i and target gap $g \in \{1, 7, 30, 120, 480\}$ days, we selected the later session j of the same task type whose gap was closest to g , within 25% of g or one day, whichever was larger. The one-day set therefore contains one- and two-day gaps, which we report separately. In monkey N, 80 training sessions were drawn at random with a fixed seed; the full monkey N set (537 pairs) also includes further gaps and was used for monitoring. All human sessions were treated as one task type. Pair counts are listed in Supplementary Table 2.

Correction ladder

Let $\mathbf{x}_j(t) \in \mathbb{R}^C$ be the raw features of test day j , $\boldsymbol{\mu}_d$ and $\boldsymbol{\sigma}_d$ the channel means and standard deviations of day d , and f_i the decoder trained on day i .

- *No correction*: $f_i((\mathbf{x}_j - \boldsymbol{\mu}_i)/\boldsymbol{\sigma}_i)$.
- *Mean updating*: $f_i((\mathbf{x}_j - \boldsymbol{\mu}_j)/\boldsymbol{\sigma}_i)$.
- *Renormalization*: $f_i(\mathbf{z}_j)$ with $\mathbf{z}_j = (\mathbf{x}_j - \boldsymbol{\mu}_j)/\boldsymbol{\sigma}_j$.
- *Re-weighting*: f_i applied to $g_c z_{j,c}$ for each channel c , with one offset per output. The weights g_c were shrunk toward 1 and the offsets toward the old intercept; the penalty was chosen from $\{10^{-6}, \dots, 10^2\}$ by five-fold cross-validation over contiguous blocks of trials.
- *Ridge to prior*: a new decoder $(\mathbf{W}, \mathbf{b})$ fitted on the lagged features $\mathbf{H}$ of day j by minimizing

$$\|\mathbf{Y} - \mathbf{H}\mathbf{W} - \mathbf{1}\mathbf{b}^\top\|_F^2 + \lambda (\|\mathbf{W} - \mathbf{W}_i\|_F^2 + \|\mathbf{b} - \mathbf{b}_i\|_2^2), \quad (1)$$

where $(\mathbf{W}_i, \mathbf{b}_i)$ is the old decoder and λ was chosen from $\{0.1, 1, \dots, 10^6\}$ by cross-validation.

- *Reference*: a decoder trained on day j 's training trials.

Supervised rungs used the first n labelled trials of day j ($n = 300$, or all training trials if fewer, unless stated). Retention was the ratio of a rung's R^2 to the reference R^2 on the same test trials, and the share of the loss recovered by re-weighting was $(R_{\text{reweight}}^2 - R_{\text{renorm}}^2)/(R_{\text{ref}}^2 - R_{\text{renorm}}^2)$. The time to half retention was found by interpolating median retention linearly in log days. For the network decoder, re-weighting was a learned per-channel gain and offset in front of the frozen network, and the refit was fine-tuning with an L_2 penalty toward the old weights; penalties were chosen on the last 20% of labelled trials. We also computed a full input remap fitted by L-BFGS⁵⁰; because it can represent any linear decoder of the same form, we used it only as a calibration target for the simulator.

Matched-output control. For monkeys C and M, sessions were rebuilt with a two-dimensional unit-vector target, the direction of hand velocity, on bins where hand speed exceeded the session's 25th percentile. The ladder was run on the same pairs.

Data efficiency in humans. For 30 training sessions per participant (fixed seed), the ladder was run with $n \in \{10, 20, 50, 100, 300\}$ labelled trials.

Decoder-weight angles. For every pair, ridge decoders ($\alpha = 10^4$) were fitted on each day. Each channel's block of weights (8 lags by all outputs) was stacked into $\mathbf{B}_i$ and

407 $\mathbf{B}_j$. We computed the angle between $\text{vec}(\mathbf{B}_i)$ and $\text{vec}(\mathbf{B}_j)$, which allows one global
 408 scale, and the angle between $\text{vec}(\mathbf{B}_j)$ and $\text{vec}(\text{diag}(\mathbf{s})\mathbf{B}_i)$, with the scale s_c of each
 409 channel fitted by least squares. As a reference, the second angle was also computed
 410 after randomly permuting the channels of $\mathbf{B}_i$.

411 Label-free corrections

412 *Procrustes rotation.* Let $\mathbf{V}_i, \mathbf{V}_j \in \mathbb{R}^{C \times k}$ hold the top $k = 16$ principal axes of the
 413 corrected training features on each day. The orthogonal matrix $\mathbf{R}$ minimizing $\|\mathbf{V}_j\mathbf{R} -$
 414 $\mathbf{V}_i\|_F$ was found by Procrustes analysis³⁴, and inputs were mapped as $(\mathbf{I} - \mathbf{V}_j\mathbf{V}_j^\top)\mathbf{z} +$
 415 $\mathbf{V}_i\mathbf{R}^\top\mathbf{V}_j^\top\mathbf{z}$. We applied it to renormalized features and, separately, to mean-updated
 416 features with $\mathbf{V}_j$ estimated from those features.

417 *CORAL.* Renormalized day- j features were re-coloured to match day i 's covari-
 418 ance, $\mathbf{z}' = \mathbf{C}_i^{1/2}\mathbf{C}_j^{-1/2}\mathbf{z}$ ³⁵, with covariances shrunk 5% toward a scaled identity.

419 *Factor-analysis stabilizer.* Following ref.²³, a factor-analysis model with 10 latent
 420 factors was fitted to each day's renormalized training features, and a ridge decoder
 421 on 8 lags of the day- i posterior latent means was trained. On day j , a model fitted
 422 without labels was aligned to day i by orthogonal Procrustes analysis of the loadings
 423 of stable channels, found by iteratively keeping the 60% of channels with the smallest
 424 loading residuals; latents were then inferred from stable channels only and passed to
 425 the frozen latent decoder. The comparison baseline reused the day- i factor model on
 426 day j .

427 Monitoring analyses

428 In monkey N, for every pair we computed Gaussian Kullback–Leibler divergences
 429 between days i and j of the projections of renormalized features on day i 's top ten
 430 principal axes (neural divergence) and of the decoder's outputs (output divergence).
 431 We report raw Spearman correlations with R^2 , partial correlations given log elapsed
 432 days and time-blocked cross-validated linear prediction of R^2 . The additional statistics
 433 are described in Supplementary Note 1. Split-half reliability of the residual after the
 434 time trend used alternating pairs of trials and the Spearman–Brown formula. For the
 435 MINDFUL data we followed the original definitions²⁷: non-overlapping 60 s windows,
 436 divergences against the first day, and the median angular error over non-excluded
 437 trials (windows with fewer than 500 valid bins skipped). We report Pearson and partial
 438 correlations given elapsed days, and leave-one-day-out linear prediction of angular
 439 error; the reference day was excluded from evaluation.

440 Regularization and recalibration

441 Ridge decoders were trained with $\alpha \in \{0.1, 10, 10^3, 10^4, 3 \times 10^4, 10^5, 3 \times 10^5, 10^6\}$ on 40
 442 training sessions per individual (22 for monkey M) and evaluated on held-out same-
 443 day trials and on later sessions at the five gaps. The 1% rule chose the largest α whose
 444 median same-day R^2 was at least 99% of the best median. We compared it with each
 445 training session's own best same-day penalty: they did not differ in monkeys C and M
 446 or in T6, and the rule added 0.004 in monkey N ($P = 6.9 \times 10^{-5}$). The tuned-penalty

ladder repeated the correction ladder on the same pairs with $\alpha = 10^4$ for both the day- i and the reference decoders. For recalibration with few trials we used 86 monkey N pairs and $n \in \{10, 20, 50, 100, 300\}$. The shrinkage λ in equation (1) was chosen per pair by cross-validation on the n trials, or as the value with the best median R^2 over pairs from other training sessions, which was 10^4 for every pair with up to 50 trials. A recalibration was harmful when its R^2 was more than 0.01 below that of the renormalized fixed decoder.

Electrode statistics

Statistics were computed per array. An electrode was active in a session when its threshold-crossing rate at -4.5 times the root-mean-square noise was at least 2 Hz. An initially active electrode (mean rate of at least 2 Hz over the first three sessions) was silent when it stayed below threshold for at least three consecutive observed sessions, and recovered when it later exceeded 4 Hz for at least three consecutive sessions. Missing sessions were skipped, not imputed. Control analyses (Supplementary Table 9) used thresholds of -3.5 and -5.5 times the noise; a fixed threshold per electrode in microvolts (its -4.5 value over the first three sessions, implemented by choosing in each session the noise multiple whose microvolt threshold was closest); removal of sessions with array-wide shifts (absolute median change in log rate above 1); a requirement for a mean waveform trough of at least $30 \mu\text{V}$; silence below 0.5 Hz; and silences lasting at least 30 days. The null distribution randomly permuted each electrode's session order. Switching rates were fitted by maximum likelihood to a two-state continuous-time Markov chain. For impedance, values of 4,000 k Ω or more were excluded and the trend was the Spearman correlation of the array median with day. For the edge effect, edge electrodes were those on the outer ring of the grid (10×10 in humans and 8×8 in monkey N), each initially active electrode's decline was the slope of $\log(\text{rate} + 0.1)$ against years, and arrays with at least 10 initially active electrodes were tested with the array as the unit. The adjusted analysis regressed slopes on edge membership and initial log rate within each array.

Simulator

The simulator transforms the z-scored features $\mathbf{Z} \in \mathbb{R}^{T \times C}$ of a real base session into those of a simulated session Δt days later (Supplementary Note 2). *Mixing*: $\mathbf{Z} \leftarrow \mathbf{Z}(\mathbf{I} + \mathbf{E})^\top$, with $\mathbf{E} = s(\Delta t)[(1 - \beta)\mathbf{G} + \beta\mathbf{L}]$, where $\mathbf{G}$ is a Gaussian matrix with zero diagonal, $\mathbf{L}$ a Gaussian matrix restricted to grid neighbours, $\beta = 0.5$ and $s(\Delta t)^2 = s_0^2 + [s_\infty(1 - e^{-\Delta t/\tau_s})]^2$. *Turnover*: each channel replaces a fraction ρ_c of its signal by a unit with rotated tuning in the session's 20-dimensional latent space; ρ_c follows a beta distribution with mean $\rho_0 + (\rho_\infty - \rho_0)(1 - e^{-\Delta t/\tau_\rho})$ and concentration 5. *Silencing*: each channel active in the base session becomes silent with probability $h_{\text{off}}/(h_{\text{off}} + h_{\text{on}})[1 - e^{-(h_{\text{off}} + h_{\text{on}})\Delta t}]$ and keeps 37% of its signal variance, the ratio of tuning strength of silent to active channels in monkey N. The switching rates ($h_{\text{off}} = 0.0038$ and $h_{\text{on}} = 0.0018$ per day) were measured from monkey N's activity in the calibration period. The six mixing and turnover parameters were calibrated on 12 base sessions before day 700 by minimizing the summed squared difference between

489 simulated and real median retention for three rungs at five gaps, with the same correc-
490 tion code and regularization as for real data, using a 24-point Sobol search⁵¹ followed
491 by Nelder–Mead⁵². Validation used 10 base sessions after day 700 and compared pre-
492 dicted and real medians for all rungs and for $n \in \{20, 50, 100\}$. Augmentation added
493 eight simulated futures per training session (Δt log-uniform between 1 and 120 days)
494 and was compared with stronger regularization, random channel dropout with gain
495 noise¹⁹ and training on the three previous sessions.

496 Statistics

497 Session pairs that share a training session are not independent. Confidence intervals
498 for medians therefore came from a cluster bootstrap over training sessions⁵³ (2,000
499 resamples, percentile intervals), and paired tests used one value per training ses-
500 sion (two-sided Wilcoxon signed-rank tests, over gaps up to 120 days for label-free
501 corrections and over all gaps otherwise). Overnight retention was compared across
502 individuals with a Kruskal–Wallis test. Harmful recalibrations were compared with
503 exact McNemar tests on paired outcomes, Holm-adjusted over the five data budgets⁵⁴.
504 Revival rates were compared with Fisher’s exact test. Electrode-edge tests used one
505 value per array (one-sided sign and Wilcoxon signed-rank tests; the direction was pre-
506 specified from earlier work³⁷). Exact P values are reported. Analyses used Python
507 3.11 with NumPy, SciPy, pandas, scikit-learn and PyTorch.

508 Use of AI tools

509 An AI assistant (Claude, Anthropic) was used to write analysis code, run analyses and
510 draft the text under the authors’ direction. The authors reviewed the code, analyses
511 and text and take full responsibility for the content.

512 Data availability

513 All data analysed in this study are public. LINK (monkey N) is available from
514 the DANDI Archive at <https://doi.org/10.48324/dandi.001201/0.251023.2336>. The
515 recordings of monkeys C and M are available at <https://doi.org/10.48324/dandi.000688/0.250122.1735>. The BrainGate yield and decoding data are available from
516 Dryad at <https://doi.org/10.5061/dryad.x0k6djj1h>. The MINDFUL data are avail-
517 able from Dryad at <https://doi.org/10.5061/dryad.n2z34tn5s>. All derived results
518 underlying the figures and tables are provided in the code repository.
519

520 Code availability

521 All code, including the simulator with its calibrated parameters, is available at <https://github.com/Imtiaj-Sajin/invasive-bci> under the MIT license. A versioned archive
522 will be deposited on Zenodo upon publication.
523

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